Biresidual Compression Network With Conditional Diffusion Model for Hyperspectral Image Compression

Jiahui Liu, Lili Zhang, Jingang Wang, and Lele Qu

***Abstract*—Hyperspectral image (HSI) compression presents the challenge of preserving both spectral and spatial fidelity while achieving high compression rates. Current compression methods frequently depend on band-by-band compression or simplistic joint modeling, which complicates the balance between spectral consistency and perceptual quality. To address this issue, a compression-driven generation framework [biresidual compression with conditional diffusion model (BRC-CDM)] is proposed, which decouples the extraction of compressed representations from the reconstruction of high-quality images. We introduce reference band information through channel-level concatenation to guide the spectral residual compression network in collaboratively extracting residual information in both spatial and spectral domains. During the prediction phase, a spectral Gaussian grid compensation (SGGC) structure is further integrated to enhance the accuracy of predictions for the target bands. Ultimately, by compressing the residual information based on the differences between the predicted bands and the true bands, an efficient representation of the residual information in the compressed domain is achieved. The predicted image is then fused with the compressed residual to obtain a more expressive and potentially compressed representation. During the decoding phase, the diffusion process of the conditional diffusion model (CDM) utilizes compressed representations as potential conditions to guide the model in progressively reconstructing images over multiple time steps. Experimental findings demonstrate that the biresidual compression (BRC) network achieves superior peak signal-to-noise ratio (PSNR) across nine datasets, with an improvement of approximately 1 dB over SOTA methods. Moreover, BRC-CDM attains a PSNR that surpasses that of the majority of current methodologies, while also delivering enhanced spectral fidelity and perceptual quality. To foster reproducibility and further development, we release the full implementation of BRC-CDM at https://github.com/Nicle-L/BRC-CDM**

***Index Terms*—Compression-driven generation, conditional diffusion model (CDM), Gaussian grid compensation, hyperspectral image (HSI) compression, residual compression.**

I. Introduction

HYPERSPECTRAL imaging has become essential in fields such as agriculture [1], disaster assessment [2], and climate research [3]. With the advancement of satellite and airborne sensing technologies [4], hyperspectral sensors now capture data at higher resolutions and wider spectral ranges.

These developments expand the application scope of hyperspectral image (HSI) but also create new challenges in data transmission and storage. Massive volumes of spectral–spatial data must be transferred from spaceborne or airborne platforms to ground stations. This increases the demand for efficient compression methods. Researchers are actively exploring new strategies to reduce data size while preserving spectral fidelity and spatial detail [5].

HSI compression faces significant challenges compared to RGB image compression. These HSIs contain a large amount of band information that results in a vast and complex data structure. Since HSIs are often used in advanced tasks such as classification [6], segmentation [7], and reconstruction [8], there is a demand for high output quality. To meet these demands, many advanced HSI compression methods have been developed. Among them, lossy compression algorithms are particularly favored by researchers for their flexibility in adjusting compression rates. Compared to learning-based methods, traditional algorithms (such as JPEG2000 [9], CCSDS [10], BPG [11], and Universal Video Coding (VVC) [12]) cannot often dynamically capture spatial features. This leads to suboptimal performance in high-dimensional data compression tasks.

Recent HSI super-resolution studies, such as MambaHSISR [13] and ConvLSTM-based methods [14], have demonstrated the effectiveness of learning-based models in modeling spatial and spectral dependencies. These studies further underscore the capabilities of learning-based networks in the representation of high-dimensional HSI. Deep learning has emerged as a dominant approach in image compression. Ballé et al. [15] pioneered the development of the first end-to-end image compression framework, which redefined conventional compression methods as a singular optimization problem. Subsequently, to improve the accuracy of image reconstruction, they introduced a hyperprior-based methodology [16] that integrated an auxiliary entropy model for the purpose of estimating the distribution of latent features. Minnen et al. [17] incorporated an autoregressive context model to enhance

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the accuracy of entropy estimation. This approach utilizes the interdependencies among previously decoded components to enhance the accuracy of predicting the distribution of each latent representation.

Although this approach increases the computational load, it significantly enhances rate–distortion performance. He et al. [18] found that the nearest neighbors contributed to a greater reduction in bit rates, while the impact of neighbors more than two elements away on bit savings was negligible. Based on this research, they proposed a parallelizable checkerboard context model aimed at reducing computational speed.

While the aforementioned methods perform well in compressing RGB images, they are less effective for HSI compression. The autoregressive context models employed in RGB image compression tend to result in degraded performance and longer decoding times when applied to HSIs. This is due to the higher level of image detail and the pronounced interspectral redundancy in HSIs, which complicates the encoder's ability to generate a structured representation of the content. Additionally, these methods primarily address the elimination of spatial redundancy and the handling of long-range pixel dependencies in RGB images, without accounting for the spectral redundancy inherent in HSI.

Current HSI compression methods frequently encounter challenges in achieving a balance between high compression ratios and the preservation of both spectral and spatial fidelity. HSIs exhibit significant interband correlations, as adjacent bands share structural patterns and spectral characteristics. However, most existing approaches depend on independent compression techniques or simplistic joint modeling. Consequently, these methods often struggle to maintain spectral consistency and perceptual quality simultaneously.

To address this issue, we propose a biresidual compression with conditional diffusion model (BRC-CDM) framework. This method integrates four key modules: spectral residual modeling, spectral Gaussian grid compensation (SGGC), residual correction compression, and a conditional diffusion model (CDM) [20]. The design aims to achieve efficient structural compression while ensuring high-quality reconstruction. In the encoding stage, the model utilizes a reference-band-guided biresidual structure to extract interband differences. It uses a Gaussian pyramid to facilitate multiscale spatial compensation for the prediction of the current band. The residual between the predicted and actual current band is subsequently compressed using a residual correction module. This process results in a compact latent representation with enhanced structural integrity. In the decoding stage, we introduce the CDM as a generative decoder. The compressed latent representation acts as a conditional input to guide the diffusion process. Through iterative denoising, the model progressively restores texture details, enabling the joint optimization of spectral dependency modeling and spatial detail reconstruction.

The main contributions of this article are as follows.

1) A novel collaborative compression-generation framework, BRC-CDM, is proposed for HSI coding. The process consists of two stages. First, structural modeling is implemented using biresidual compression (BRC) network. Spectral backbone features undergo quantized residual encoding and entropy coding to produce compact latent representations. The second stage involves image reconstruction with a CDM serving as the generative decoder. The latent representations act as conditioning inputs for the diffusion sampling path. The CDM employs iterative noise injection and denoising techniques to recover high-frequency spatial details and texture information. It has broken through the limitations of traditional compression methods in spectral redundancy modeling and detail restoration, successfully achieving an optimized balance among compression efficiency, spectral fidelity, and perceptual quality.
2) A spectral residual compression network is designed to efficiently model spectral redundancy and structural differences. Current and reference bands are concatenated along the channel dimension. A spectral residual learning mechanism models and compresses spatial–spectral differences. Only the changing regions are encoded, which enhances compression efficiency and maintains spectral consistency.
3) An SGGC structure and a residual correction compression network are introduced for multiscale prediction and residual adaptation. Inspired by the concept of hierarchical representation in neural volume rendering, the SGGC module constructs a multiscale Gaussian pyramid for the prediction of current band-oriented structures and the compensation of the reference band. The residuals between the predicted and actual bands are nonlinearly encoded using a deep network, creating an efficient prediction-residual compression pipeline.
4) Experiments on nine datasets have confirmed the effectiveness of BRC-CDM. This method achieves talented compression efficiency while ensuring image quality. Compared to SOTA methods, it demonstrates exceptional performance with regard to spectral fidelity and perceptual metrics.

The organization of the subsequent sections is delineated as follows. Section II offers a review of contemporary research pertaining to HSI compression and examines the recent applications of diffusion models. Section III delivers an in-depth account of the design and implementation of the BRC-CDM method. Section IV presents the experimental results, which is complemented by a comprehensive assessment of the efficacy and advantages of the proposed methodology. Section V addresses the limitations of the model and discusses practical deployment challenges. Finally, Section VI concludes the study.

II. Related Work

This section first addresses the specific challenges associated with HSI and then outlines various compression methodologies and solutions relevant to this field. Finally, a review of contemporary applications of diffusion models is presented.

A. Neural Network-Based HSI Compression

HSI captures information across multiple spectral bands. This provides much richer spectral data than RGB images.

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However, due to the typically large volume of HSI data, traditional compression methods often struggle to handle it. Consequently, deep learning-based approaches for HSI compression have been developed to address these challenges. In HSI compression, researchers face two primary challenges:

First, HSIs contain hundreds of spectral bands, far exceeding the three channels of standard RGB images. This high spectral resolution significantly increases data volume, which imposes substantial demands on storage and transmission systems.

Second, it is essential to preserve both the spectral and spatial characteristics of the image during compression. There is a complex correlation between the spectral channels and spatial dimensions of hyperspectral data, which requires precise maintenance of this relationship during compression. Inadequate management may lead to the loss of critical spectral information in the decompressed image, thereby adversely affecting subsequent analyses and applications. Conventional compression algorithms, even when applied to a small number of bands such as JPEG [21] and JPEG2000 [9], are unable to efficiently exploit the redundancy between spectral and spatial information in HSI. This leads to less efficient processing of such data.

Guo et al. [22] proposed an edge-guided HSI compression network that leverages edge information within HSI to achieve high-quality reconstruction. By capturing edge features, this method significantly enhances the preservation of structural details and spatial information in the reconstructed images. However, it falls short in extracting interspectral information among different bands of HSI. In contrast, Zhang et al. [23] introduced a fast neural radiation field strategy for HSI compression. Rather than relying on conventional transform-based techniques, their method encodes hyperspectral coordinate information through a multilayer perceptron. This approach offers greater flexibility in managing complex hyperspectral features. This leads to improved compression efficiency and reconstruction quality. However, the method mainly addresses redundancy between hyperspectral pixels and is less efficient at reducing redundancy between spectral bands.

Mijares i Verdú et al. [24] investigated a scalable and low-complexity compression method for HSI. When processing K input bands, the number of hidden layers N and latent layers M of the filters in the architecture should follow a linear relationship with K: $N = 32(K + 1)$ and $M = 128K$. This approach ensures the effectiveness and applicability of the method as well as reduces the complexity of the model, but it also does not provide a reasonable treatment of interspectral redundancy. Zhao et al. [25] approached HSIs as sequences of frames, which facilitates the capture of spectral dynamics. This method neglects global spectral dependencies and compromises spatial–spectral coherence by segmenting images into individual frames. Fang et al. [26] introduced a three-stage hybrid compression framework with a modular design. However, the stage-wise training limits its performance.

B. Recent Applications of Diffusion Models in Image Compression

In recent years, CDMs have gained considerable attention for their ability to incorporate conditional information during the generation process. Traditional generative models generate data unconditionally, with no control over specific output characteristics. In contrast, CDMs allow for more precise generation. They achieve this by building conditions or constraints into the process.

Nichol et al. [27] introduced contextual information into CDMs, which allowed the model to generate images that matched a given textual description. Similarly, Rombach et al. [28] achieved notable success in multimodal generation tasks by using CDMs to produce high-quality images based on textual inputs. These advancements highlight CDMs' potential in various applications, including image generation, inpainting, editing, and multimodal generation.

Yang and Mandt [29] first introduced diffusion models into the field of RGB image compression. This approach pioneers the use of diffusion-based generation in compression. However, it encounters training challenges common to large models, particularly in balancing multiple loss terms. As a result, optimization becomes difficult, training time increases, and performance gains remain limited. After evaluating and integrating the strengths and weaknesses of previous studies, we use the CDM to generate high-quality images.

III. Methodology

In this section, the proposed BRC-CDM and its two innovative design features will be discussed in detail. The discourse commences with a comprehensive examination of BRC-CDM, subsequently introducing two innovative designs: a BRC network and a reconstruction methodology grounded in CDM.

A. Overall Framework

The proposed BRC-CDM network is presented in Fig. 1. The BRC-CDM framework consists of four primary components: the spectral residual compression network, the SGGC structure, the residual correction compression network, and the CDM.

HSIs contain rich spectral information and exhibit strong interband correlations, as adjacent bands often share similar spatial structures and spectral responses. Compressing each band independently overlooks this inherent redundancy, resulting in suboptimal encoding efficiency.

In order to tackle this problem, we put forward a spectral residual compression network. In this compression framework, the compression process for HSI involves two input images: the current band image $x_{\text{cur}} \in R^{W\times H\times 3}$ and the reference band image $x_{\text{ref}} \in R^{W\times H\times 3}$. These inputs have consistent spatial dimensions but differ in band selection. In the proposed framework, the reference image is composed of the first three bands (bands 0, 1, and 2), while the current band image consists of the three bands (bands 3, 4, and 5). In this approach, the bands of the HSI are systematically divided into several groups. Each group consists of three consecutive bands, which are then processed sequentially.

The input to the network is a six-channel tensor. It is formed by concatenating the current band with a reference band along the channel axis. This structure enables joint modeling of spectral and spatial dependencies. The network encodes

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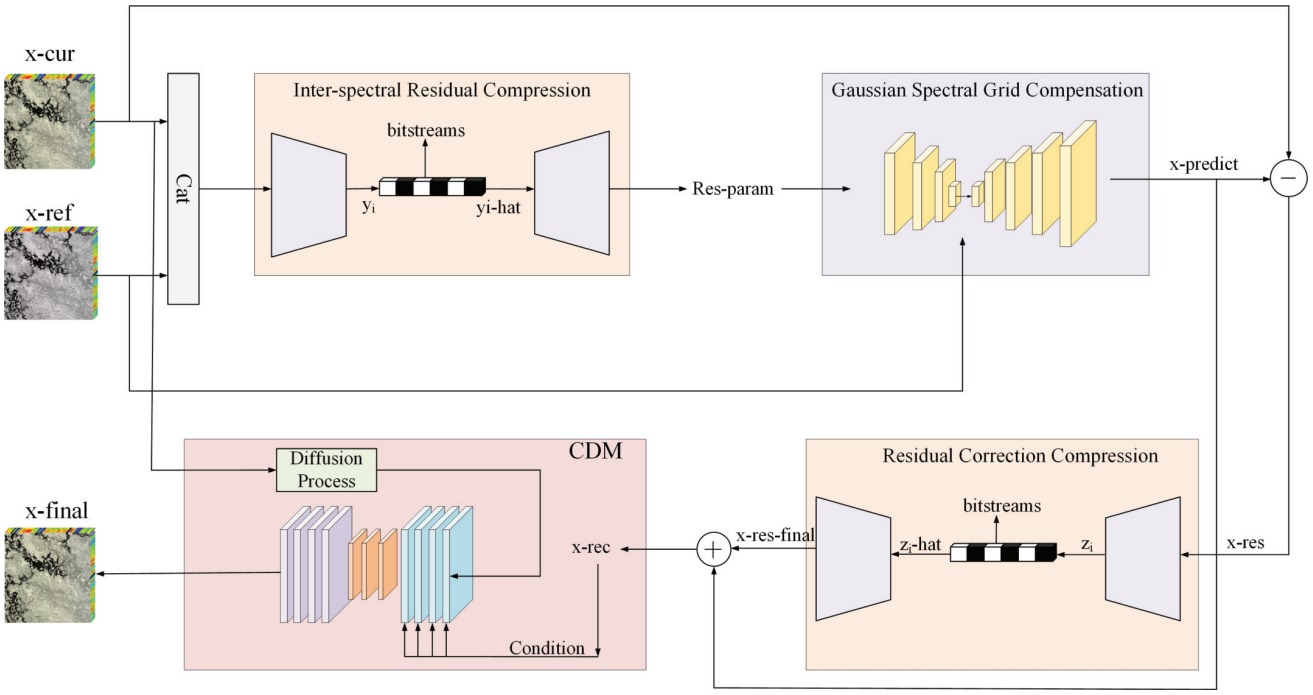


Fig. 1. Overall BRC-CDM network. Here, y_i denotes the latent representation extracted by the interband residual compression network. $\hat{y}_i$ represents the reconstructed latent feature obtained by decoding the quantized bitstream using an AD, which serves as the input to the decoder module. Similarly, z_i denotes the latent representation produced by the residual correction compression network, and $\hat{z}_i$ refers to its decoded version, also recovered via arithmetic decoding and fed into the corresponding decoder. The spectral residual compression network encodes the residuals between the current and reference bands, producing a three-channel residual map. The SGGC module utilizes spatial–spectral residuals to adjust the reference band and predict the current band. The residual correction network compresses the difference. The compressed representation of the current band serves as a latent condition to guide the sampling trajectory in the CDM.

the interband residuals and outputs a three-channel residual map. The residual map captures localized variations, such as texture differences and luminance shifts. The network does not reconstruct the full-band content but focuses on encoding meaningful residual information.

The SGGC structure, based on the residual map, enables accurate prediction of reference bands. Inspired by differential pulse code modulation (DPCM) [30], the residual correction compression network is introduced for improved compression. In this context, the interspectral residual compression network and the SGGC structure in the first stage collectively function as the predictor; they utilize the reference band information to generate preliminary predictions for the current band. Meanwhile, the residual correction compression network in the second stage operates similarly to the "residual encoder" in DPCM, specifically focusing on the fine-grained compression of the residuals left by the predictions.

To enhance reconstruction fidelity, a CDM is used as a generative decoder. The compressed residual representation derived from the biresidual pipeline functions as a latent conditional input, thereby directing the process. Commencing with Gaussian noise, the CDM performs iterative denoising toward the target image. The conditional context preserves spectral consistency and supports the recovery of high-frequency details, including edges and fine textures.

B. Spectral Residual Compression Network

Based on the concept of flow network [31], an end-to-end spectral difference extraction network is designed to efficiently capture and compress subtle variations between different bands. The specific structure is shown in Fig. 2.

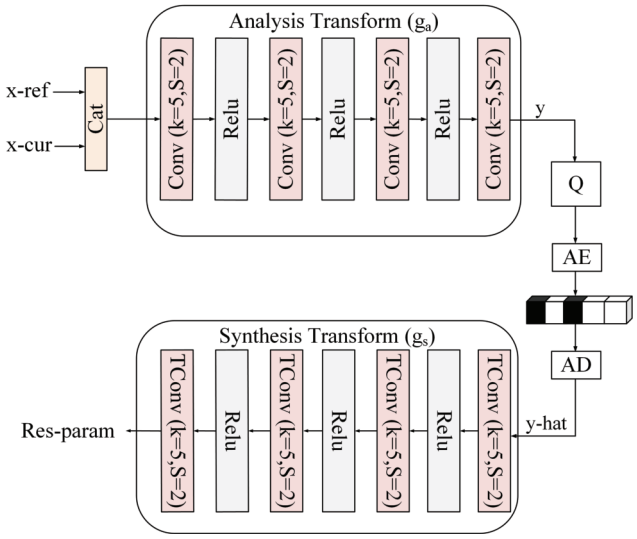


Fig. 2. Structure of the spectral residual compression network. AE stands for Arithmetic Encoder, AD stands for Arithmetic Decoder, and Q represents the Quantizer.

The primary aim of the present compression network is to effectively identify and encode the variances in information across the various bands. This approach exploits the potential differences between the bands. It also improves the efficiency and accuracy of subsequent image compression. In detail, by concatenating the input bands and the reference bands along

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the channel dimension, this concatenated tensor is used as input to the first compression network and fed to the encoder. During the decoding phase, the decoded image is obtained through optimization with a regional differential loss. This process leverages feature encoding techniques to deeply model the potential differences between bands.

When using the encoder g_a is designed to represent the latent variables y compactly and efficiently. The latent variables y are then quantized using the quantization function Q, which results in the quantized representation $\hat{y}$. After quantization, $\hat{y}$ becomes a discrete value, which can be entropy-coded and transmitted as a bitstream. Quantization introduces errors that lead to an optimization problem of the rate–distortion. Finally, by combining the quantized representation $\hat{y}$ with the decoder, the decoded residual parameters are generated. This process can be described as follows:

$$y = g_a\left(\text{cat}\left(x_{\text{cur}}, x_{\text{ref}}\right); \phi_g\right) \tag{1}$$

$$\hat{y} = Q\left(y\right) \tag{2}$$

$$\text{Res} = g_s\left(\hat{y}; \theta_g\right). \tag{3}$$

In this model, the encoder g_a and decoder g_s are both constructed using deep neural networks, where their trainable parameters are denoted by ϕ_g and θ_g, respectively. Since the quantization operation produces near-zero gradients concerning ϕ_g, the quantizer Q adopts a noise-based approximation during training. Specifically, uniform noise in the range [0.5,0.5] is added to the latent features to simulate discretization. This enables gradient-based optimization. During inference, actual quantization is applied using rounding.

The quantized features are subsequently processed by the arithmetic encoder (AE), which performs entropy coding. The AE assigns shorter code words to symbols with higher probabilities, thereby reducing the bit rate and improving compression efficiency. A nonadaptive binary arithmetic coding method is utilized. On the decoder side, the arithmetic decoder (AD) reconstructs the original quantized features from the compressed bitstream.

C. Spectral Gaussian Grid Compensation

To mitigate model complexity and enhance the precision of structure-aware compensation, we introduce an innovative SGGC architecture for predicting current bands. The SGGC framework retains the ability for multiscale perception while simultaneously reducing the number of trainable parameters and the associated computational burden. Drawing inspiration from neural volumetric rendering methodologies, this approach separates spectral alignment from dense prediction by implementing a hierarchical Gaussian field construction and continuous coordinate warping.

SGGC comprises two major components: 1) the Gaussian grid construction structure and 2) the flow-guided spatial warping module.

1) Gaussian Grid Construction Structure: This module hierarchically encodes the reference band by sequentially applying Gaussian filters and average pooling. Each scale level is restored to the native resolution via bilinear upsampling. The resulting multiscale features are stacked along the channel dimension, producing a dense spectral volume $G \in R^{W\times H\times 3}$ that retains both low- and high-frequency structure information. The formula is shown below

$$G = \text{Stack}\left\{\text{Up}\left(\text{AvgPool}\left(G_\sigma\left(R\right)\right)\right) | \sigma \in \Sigma\right\} \tag{4}$$

where R is the reference band. G is a Gaussian filter with scale σ. Up$(\cdot)$ denotes bilinear interpolation. And Stack$(\cdot)$ represents a feature stacking operation on the channel dimension.

2) Flow-Guided Spatial Warping Module: To achieve precise alignment and structure-aware compensation between the current and reference spectral bands, we introduce a flow-guided spatial warping module composed of two flow fields: a spectral shift map and a local response map.

3) Spectral Shift Map: The spectral shift is modeled as a 3-channel vector $F_s \in R^3$, applied per spectral channel to the reference band. This map captures coarse structural differences in the spectral dimension by applying channel-wise shifts

$$\hat{I_{\text{ref}}}(x, y, c) = I_{\text{ref}}\left(x, y, c + F_s\left(c\right)\right). \tag{5}$$

In this context, $I_{\text{ref}}(x, y, c)$ represents the intensity of the reference band at the spatial coordinates (x, y) and the spectral channel c. Additionally, $F_s(c)$ signifies the offset associated with the cth channel.

a) Local response map: To account for spatial misalignments and local deformations, we define a flow field $F_l \in R^{H\times W\times 2}$, representing per-pixel displacements

$$\left(x', y'\right) = (x, y) + F_l\left(x, y\right). \tag{6}$$

These warped coordinates (x', y') are used to sample the spectrally shifted reference band using bilinear interpolation

$$\tilde{\mathbf{I}}_{\text{ref}}\left(x, y, c\right) = \text{BilinearInterp}\left(\hat{\mathbf{I}}_{\text{ref}}, x', y', c\right) \tag{7}$$

where BilinearInterp$(\cdot)$ denotes a differentiable 2-D interpolation function over spatial coordinates.

Finally, we apply 3-D interpolation over the Gaussian grid to sample the warped coordinates. This concludes the implementation of structure-aware compensation. Importantly, the entire warping process requires no trainable parameters. The complete structure-aware warping procedure is formulated as follows:

$$\tilde{I}_{\text{ref}} = \text{Warp}_{\text{space}}\left(\text{Shift}_{\text{spectral}}\left(I_{\text{ref}}, F_s\right), F_l\right) \tag{8}$$

where $\text{Shift}_{\text{spectral}}(\cdot)$ performs spectral channel shifting and $\text{Warp}_{\text{space}}(\cdot)$ performs spatial deformation via coordinate warping.

D. Residual Correction Compression Network

Due to the fact that the prediction process is based on transformations derived from the reference image, and the input image is only used during encoding to extract band differences, all information is processed based on the reference band. This can affect the accuracy of the prediction process. So we subtract the predicted image obtained through transformation from the actual input image to obtain the true residual. The true residual is then processed through the same encoding and decoding network. The decoded residual is incorporated into the predicted image to produce the final reconstructed image.

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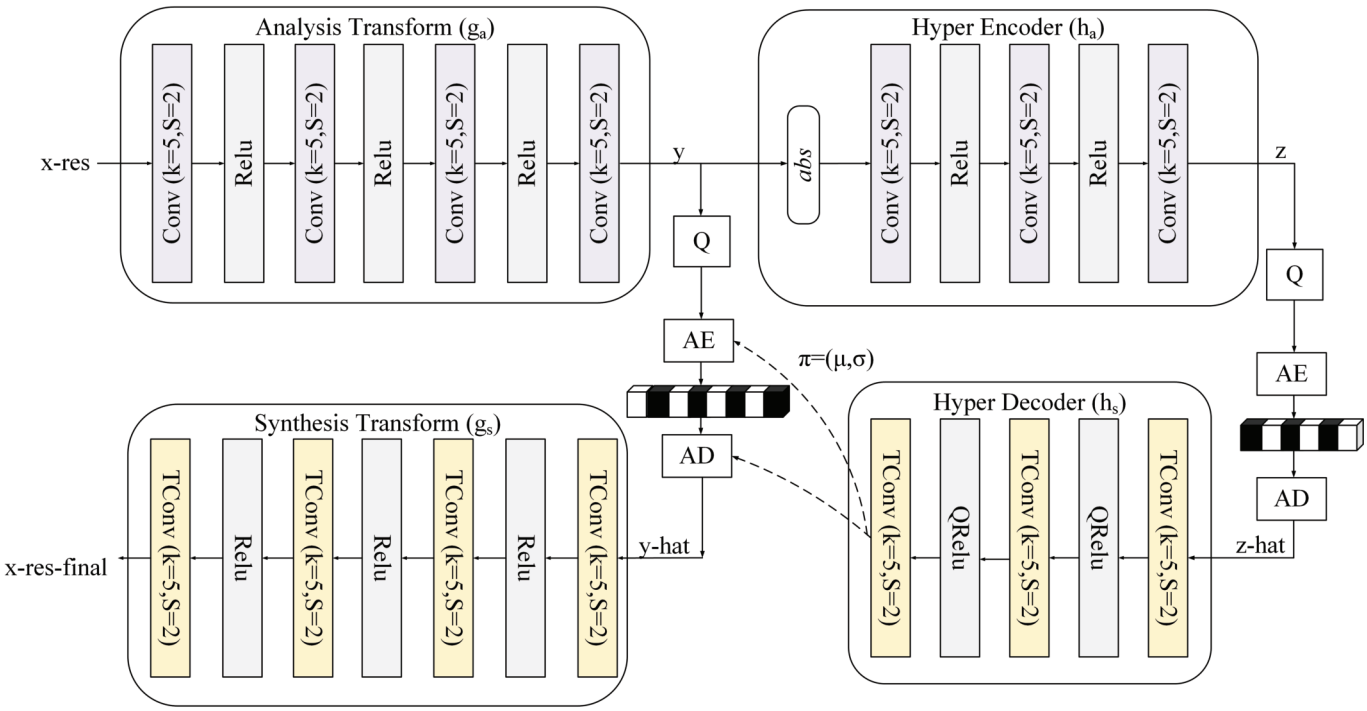


Fig. 3. Structure of the residual correction compression network.

By feeding the residuals between predicted and actual values into a residual correction compression network, residual information is efficiently encoded within the compressed domain. This approach stores only critical residual components. It does not store full image data. This optimizes storage efficiency. To implement this, a structurally simple compression network is designed to encode residuals. The detailed network architecture is illustrated in Fig. 3.

Similar to other compression methods [16], [19], [35], side information z is introduced to capture the spatial dependencies y through hyperencoder (h_a). The side information z corresponds to the prior of the latent representation, which serves both as a prior for the entropy model parameters and as a hyperprior for the latent representation. Through the quantization process, the quantized information $\hat{z}$ is precisely obtained. The probability distribution is then efficiently estimated using the hyperdecoder h_s, which performs $\hat{z}$ conditional inference to further optimize compression performance. It improves overall data compression performance by introducing an advanced probabilistic model that captures and exploits the high-order statistical properties of the latent representation. This process can be described as follows:

$$z = h_a(y; \phi_h) \tag{9}$$

$$\hat{z} = Q(z) \tag{10}$$

$$\text{Mean, Scale} = h_s(\hat{z}; \theta_h). \tag{11}$$

In this framework, the symbols h_a and h_s represent the hyperencoder and hyperdecoder networks, which are parameterized by ϕ_h and θ_h, respectively. The entropy model is responsible for estimating the distribution parameters $\pi = (\mu, \sigma)$ associated with a Gaussian prior. For mean prediction, a standard hyperdecoder with ReLU activation is employed, while variance prediction utilizes a QReLU-activated decoder. The adoption of QReLU [32] is essential for scale estimation, as it introduces quantization and a clipping mechanism to constrain output values within a valid range. This ensures numerical stability and facilitates precise bit rate control. By incorporating deep feature representations and nonlinear mappings, the model effectively captures interband spectral dependencies and enhances compression efficiency during encoding.

For the architecture details of the residual correction compression network (as shown in Fig. 3), "Conv | $k5$ | $s2$" denotes a convolutional layer with a kernel size of 5×5 and a stride of 2, which is used for downsampling, while "TConv" refers to a transposed convolution layer (ConvTranspose2d) used for upsampling. Our hyperprior structure is based on the literature [16]. The encoder consists of four downsampling convolutional layers, while the decoder is composed of four upsampling convolutional layers. The hyperencoder includes three downsampling convolutional layers, and the corresponding hyperdecoder utilizes three upsampling convolutional layers.

Given the use of two compression networks, their loss functions can be defined as follows:

$$\begin{aligned}\mathcal{L}_{\text{Bi-res-loss}} &= R_{\text{res1}} + R_{\text{res2}} + \lambda \cdot D \\ &= \mathbb{E}_{x\sim p_x}\left[-\log_2 p_{\hat{y}}\left(Q\left(g_a\left(x_{\text{input}}\right)\right)\right)\right] \\ &\quad + \mathbb{E}_{x\sim p_x}\left[-\log_2 p_{\hat{y}}\left(Q\left(g_s\left(x_{\text{res}}\right)\right)\right)\right] \\ &\quad + \lambda \cdot d\left(x_{\text{cur}}, \hat{x}_{\text{cur}}\right).\end{aligned} \tag{12}$$

In the model, x_{input} is defined as the concatenation of two key components: x_{cur}, representing the current input, and x_{ref},

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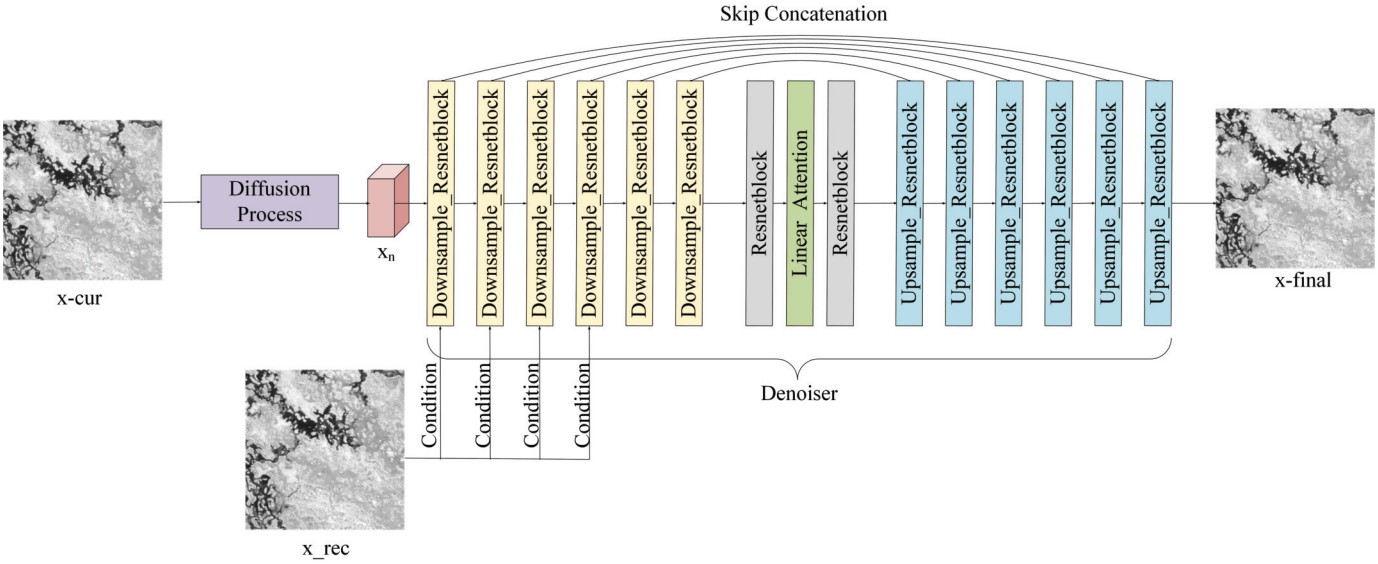


Fig. 4. Structure of the CDM.

which acts as the reference input, combined in the channel dimension. This configuration allows the model to process spectral or spatial relationships between the current and reference inputs in a unified manner. The overall compression process involves two distinct residual compression networks. The total rate is the combination of these two networks, where R_{res1} captures the rate from the spectral residual compression network, and R_{res2} accounts for the rate from the residual correction compression network. These rates are indicative of the bit costs necessary for encoding the residuals. The distortion $d(x_{\text{cur}}, \hat{x}_{\text{cur}})$ is measured as the MSE between the input bands and their decoded counterparts, which emphasizes the accuracy of the reconstruction. This balance between rate and distortion defines the efficiency of the compression process.

E. Conditional Diffusion Model

Within the diffusion framework, Gaussian noise is incrementally introduced to the image across several temporal steps during the forward process. The reverse process iteratively denoises this noisy signal to reconstruct the original image. These two stages form a denoising diffusion probabilistic model (DDPM), enabling robust reconstruction under varying noise levels. The architecture is illustrated in Fig. 4.

1) Forward Diffusion Process: Given an input image x, Gaussian noise is added over t time steps

$$x_t = \sqrt{\alpha_t} x_{t-1} + \sqrt{1-\alpha_t} z_t \tag{13}$$

where $z_t \sim \mathcal{N}(0,1)$ and $\alpha_t = 1 - \beta_t$. A closed-form expression relates x_t to x_0

$$q(x_t|x_0) = \mathcal{N}\left(x_t;\ \sqrt{\bar{\alpha}_t} x_0, (1-\bar{\alpha}_t) I\right) \tag{14}$$

with $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$.

2) Reverse Denoising Process: The model learns to reverse the noise process using a parameterized distribution

$$p(x_{t-1}|x_t) = \mathcal{N}\left(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)\right). \tag{15}$$

The training objective minimizes the prediction error between real and estimated noise

$$\mathcal{L}_{\text{denoise}} = \| \epsilon_t - \epsilon_\theta(x_t, t) \|_2^2 . \tag{16}$$

A U-Net backbone is used to predict ϵ_θ. It includes four downsampling and upsampling stages, with ResNet blocks and linear attention to capture both spatial detail and long-range dependencies. Skip connections help preserve structure across scales.

To accelerate training, the BRC network is pretrained and frozen. The diffusion model is trained separately following the consistency training strategy in [42].

The CDM enhances spectral modeling by explicitly incorporating interband correlation into the generative decoding process. In the proposed BRC-CDM framework, the compressed representation produced by the BRC network preserves the main spectral structures and variations between adjacent bands. This representation is treated as a latent spectral condition and is fed into the CDM as a guiding prior.

During reverse diffusion, the model iteratively denoises Gaussian noise under the control of this condition. At each step, the spectral prior enforces consistency between the generated output and the spectral distribution of the original hyperspectral signal. The CDM does not treat bands independently. Instead, it uses the learned spectral representation to constrain the generation process, ensuring that spectral transitions remain smooth and physically plausible.

By integrating this spectral prior, CDM maintains spectral fidelity across all bands. It prevents shifts or discontinuities that may arise from hallucinated details. At the same time, its multistep sampling mechanism enables the gradual recovery of spatial texture. This joint spectral–spatial enhancement

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distinguishes CDM from traditional decoders, which often fail to explicitly capture high-order spectral dependencies.

3) CDM Loss Function: The final loss includes two terms: noise prediction loss and reconstruction loss

$$\mathcal{L}_{\text{cdm}} = \text{MSE}\left(\epsilon_t, \epsilon_\theta\left(x_t, t\right)\right) + \text{MSE}\left(x, \hat{x}\right) \tag{17}$$

where x is the input band and $\hat{x}$ is the reconstructed output. The second term can be replaced by a perceptual loss if required.

F. Total Loss Function

The overall training objective comprises two main components. The first corresponds to the loss used for optimizing the BRC network, which guides bitstream generation during compression. The second is the loss function for the diffusion model, which includes two terms: the mean squared error (MSE) between the injected noise and the noise predicted by the model, and the MSE between the original image and the final reconstruction obtained through the denoising process. The complete formulation of the loss function is given as follows:

$$\begin{aligned} L_{\text{total}} &= R + \text{MSE}\left(\epsilon_t, \epsilon_\theta\left(x_t, t\right)\right) + \lambda \cdot \text{MSE}\left(x, \hat{x}\right) \\ R &= R_{\text{res1}} + R_{\text{res2}} \\ &= E_{x\sim p_x}\left[-\log_2 p_{\hat{y}}\left(Q\left(g_\alpha\left(x\right)\right)\right)\right] \\ &\quad + E_{x\sim p_x}\left[-\log_2 p_{\hat{y}}\left(Q\left(g_a\left(x_{\text{res}}\right)\right)\right)\right]. \end{aligned} \tag{18}$$

In this context, x represents the input band. x_{res} denotes the difference. During the inference phase, the trained compression model is used with its parameters frozen, and the BRC compression network compresses the input band.

IV. Experimental Results and Analysis

In order to assess the efficacy of BRC-CDM, a comparative analysis is conducted against various HSI compression methods, including PCA+Jpeg2000 [34], Verdú (2023) [24], FHNeRF (2024) [23], JSCIFI-F(2025) [25], and ARM-Net(2025) [26]. Additionally, we compared BRC-CDM with several RGB image compression methods, such as JPEG2000 [9], BPG [11], Context (2018) [17], Cheng (2020) [35], and the latest diffusion model-based image compression method, CDC (2023) [29], to provide a comprehensive performance analysis.

A. Experimental Configurations

1) Dataset: We evaluated the BRC-CDM algorithm using nine publicly available HSI datasets, including Chikusei, Botswana, Pavia (covering both Pavia Center and Pavia University), Houston (DFC 2013 and DFC 2018), KSC, Washington dc, and a dataset from Xiong'an, China. These tests comprehensively demonstrate the potential and prospects of BRC-CDM in the context of HSI compression performance.

2) Evaluation Metrics: Compression efficiency is measured using bits per pixel per band (bpppb). To evaluate reconstruction quality, four complementary metrics are adopted: PSNR for pixel-wise fidelity [36], MS-SSIM for multiscale structural similarity [37], LPIPS for perceptual similarity based on deep features [38], and SAM for assessing spectral accuracy [39].

TABLE I
Impact of Different Number of Bands on the Model

Bands num	Flops (G)	FPS	Parameters (M)
Bands = 3	27.88	51.04	23.83
Bands = 5	109.49	22.71	500.14
Bands = 7	151.24	10.96	750.35

3) Experimental Settings: This study uses nine benchmark HSI datasets. Each image is resized to 256 × 256 pixels and divided into train and test sets, with 20% of the data reserved for testing. All methodologies based on deep learning are executed utilizing the PyTorch framework and are trained on a singular NVIDIA RTX 4090 GPU. The Adam optimizer [40] is employed for this purpose. To ensure a fair comparison, all models are subjected to the same training environment and hardware configuration. The learning rate is initialized following previous work [31] and adjusted according to performance. To compare RD performance across bit rates, we use five λ values: {512, 2048, 4096, 8192, 32678}. The compression model adopts $N = 128$ and $M = 192$ as its latent dimensions. Traditional codecs are also evaluated. JPEG2000 is implemented via the OpenJPEG library, and BPG is executed using a public C-language implementation. All evaluations are performed under identical runtime conditions.

During the training of the diffusion model, the parameters of the compression network are frozen, and a gradually decreasing learning rate strategy is used for optimization. For the first 2000 epochs, the initial learning rate is set to $1e^{-4}$ and remains constant. The learning rate is then reduced by a quarter every 1000 epochs until it reaches $1e^{-6}$. It then remains constant. The entire training process is monitored and controlled. This ensures that the model learns and adjusts its parameters at various stages. As a result, the model achieves optimal performance in both compression and reconstruction.

B. Band Selection Strategy

By calculating the PSNR between adjacent bands, a high correlation is observed. Based on this observation, the dataset is structured around band grouping, where every three adjacent bands form a group. For example, the first group consists of bands 0, 1, and 2, while the second group consists of bands 3, 4, and 5, and so on. We will then explain why we have chosen three bands as a group.

Comparative experiments are performed with different numbers of input bands. In particular, groups of 3, 5, and 7 spectral bands are tested. The findings from the experiments are displayed in Table I. Instances with fewer than three input bands are omitted due to the heightened computational demands associated with their processing. After evaluating model complexity and parameter size, it is found that the use of three bands provides an optimal balance. This choice simplifies the model structure, avoids unnecessary complexity in the diffusion model inputs, and reduces the computational burden during postprocessing.

The experimental findings indicate that an increase in the number of input bands substantially elevates the computational demands of the model. This is attributed to the fact that a

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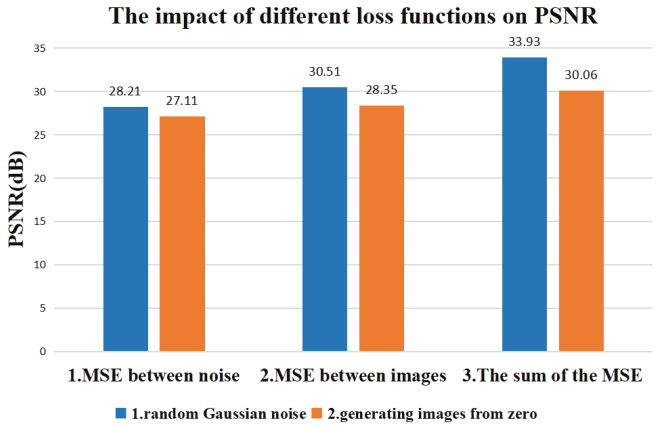


Fig. 5. Comparison of results using six different loss functions at 0.35 bpppb.

greater number of bands leads to an escalation in the dimensionality of the input data. This requires a greater number of channels in the network layers. While using a single band may reduce the number of parameters, it ultimately increases computation time and training costs. The use of three bands strikes a balance between preserving spectral information and managing computational efficiency and model complexity. This makes it a more practical choice.

C. Analysis of Results With Different Loss Functions

In diffusion models, two approaches are employed for the processing of input images: the first involves the incorporation of random Gaussian noise into the images, while the second entails the generation of images from an initial state of randomness. Correspondingly, there are three optimization directions for the loss functions of the diffusion model: The initial component is the MSE loss associated with the noise elements. The subsequent component pertains to the MSE loss calculated between the input and output images. The final component represents a synthesis of both aforementioned losses.

In comparative experiments, these methods are tested independently while applying the same learning rate tuning strategy. Each method ran 10 000 experiments, with a total of approximately 60 000 iterations across all methods (using NVIDIA 4090 GPU). A total of six different combinations of loss function configurations are evaluated, and the findings are presented in Fig. 5. Ultimately, the configuration of the loss function that demonstrates the highest performance is designated as the benchmark for future experimental procedures. An in-depth study of loss functions for diffusion models is performed, and the convergence speed is significantly accelerated by optimizing these loss functions.

Experimental results indicate that under the same learning rate decay strategy, the diffusion model produces noticeable quality degradation at low bit rates. At higher bit rates, this degradation becomes significantly less pronounced. The performance drop at low rates is attributed to insufficient conditional information, which constrains the generative ability of the model. When richer conditioning is available, the selected loss function effectively mitigates reconstruction errors.

TABLE II

BD-RATE (%), BD-PSNR (DB), AND BD-MS-SSIM (DB) PERFORMANCE COMPARISON ON THE HSI DATASET

Method	BD-Rate (%)	BD-PSNR (dB)	BD-MS-SSIM (dB)
PCA+jpeg2000 [34]	−0.70%	−3.98	+1.31
BPG [11]	−0.70%	+1.15	−1.57
Context (2018) [17]	+0.81%	+4.68	+7.66
Cheng (2020) [35]	+0.41%	+4.85	+7.93
CDC-0.0 (2023) [29]	−0.65%	−7.51	+0.16
CDC-0.9 (2023) [29]	−0.23%	−9.07	−2.41
Verdú (2023) [24]	+1.62%	+4.16	+7.61
FHNeRF (2024) [23]	−0.55%	+5.52	+8.71
JSCIFI-F (2025) [25]	+0.32%	+5.66	+8.56
ARM-Net (2025) [26]	+0.28%	+5.43	+8.92
BRC (Ours)	−0.19%	+5.77	+9.34
BRC-CDM (Ours)	−0.19%	+3.87	+6.72

D. Performance on Four Evaluation Indicators

1) Performance on Rate Distortion: The BRC and BRC-CDM models consistently outperform conventional compression algorithms in terms of both index across different rates. Although the rate–distortion curve is not specifically optimized for the MS-SSIM metric, the BRC model still demonstrates competitive performance, particularly at higher bit rates. The BRC-CDM model shows slightly lower MS-SSIM scores compared to some deep learning-based techniques; however, it surpasses traditional codecs and offers significant advantages over diffusion-based methods, such as CDC [29].

As illustrated in Table II, we evaluate the rate–distortion performance of BRC-CDM in comparison to various baseline methods using three established metrics: BD-Rate, BD-PSNR, and BD-MS-SSIM, with JPEG2000 serving as the reference codec. These metrics provide a comprehensive assessment of bit rate efficiency, reconstruction quality, and perceptual fidelity.

BD-Rate measures the bit rate difference required to achieve the same quality. It is expressed as a percentage. A lower (negative) value indicates better compression efficiency. BD-PSNR quantifies the image quality in dB under a fixed bit rate. A higher value indicates superior reconstruction. BD-MS-SSIM reflects the perceptual quality based on the MS-SSIM index. A higher score indicates stronger visual consistency. Both BRC-CDM and BRC achieve significant bit rate savings across all nine datasets. The BRC model also achieves the highest gains in PSNR and MS-SSIM, demonstrating superior reconstruction performance.

As illustrated in Fig. 6, the proposed BRC compression model exhibits outstanding rate–distortion performance across different rates. Furthermore, the proposed BRC-CDM outperforms most methods at high bit rates.

2) Performance on Other Indicators: Fig. 7 illustrates the performance of each method in terms of SAM and LPIPS. For the SAM metric, both BRC and BRC-CDM achieve notably better spectral fidelity than existing deep learning-based compression algorithms. This advantage becomes especially clear under the same bit rate condition. Compared to other models, which tend to suffer from spectral distortion, BRC and BRC-CDM maintain lower spectral error and better interband consistency. The results indicate that both models are more effective in preserving spectral integrity and are more stable across varying compression settings.

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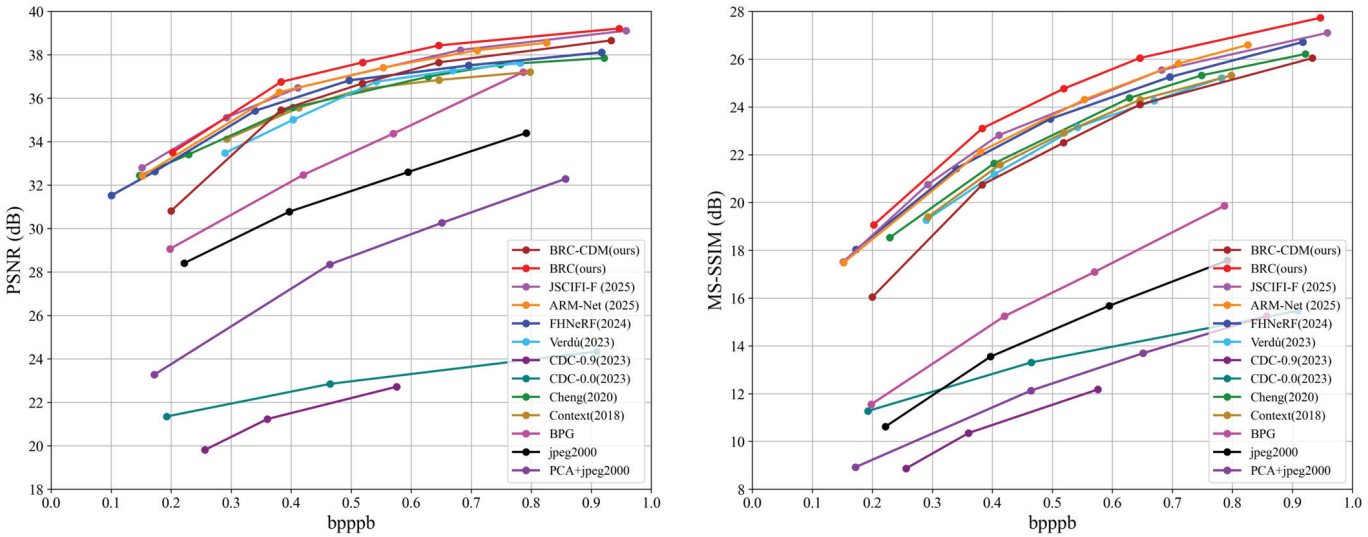


Fig. 6. Comparison of rate–distortion performance.

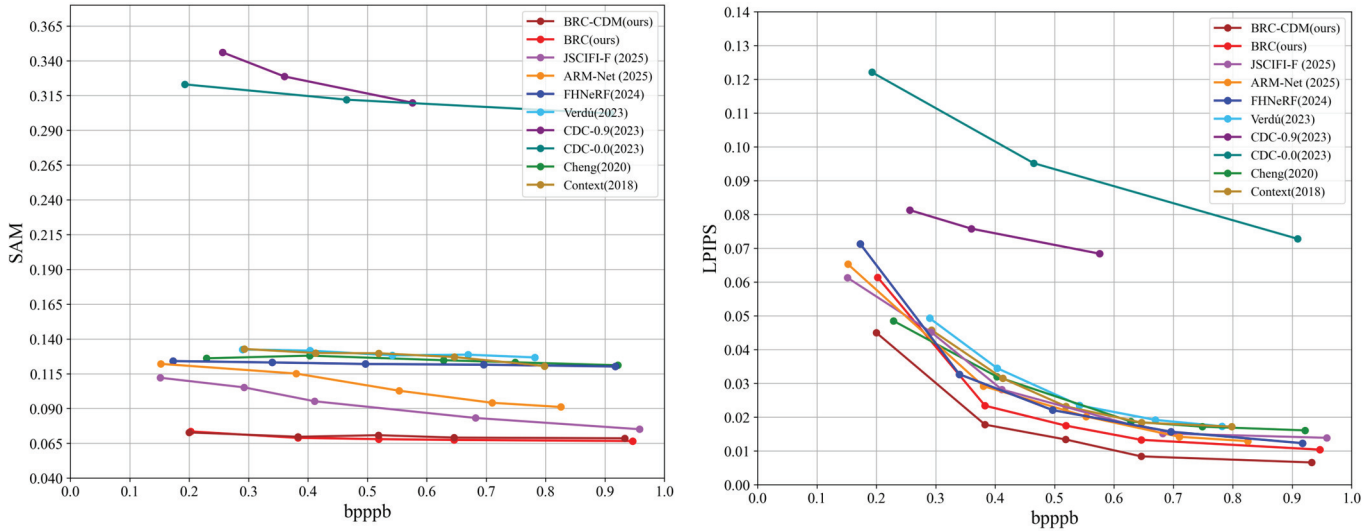


Fig. 7. Comparison of spectral consistency and perceptual quality.

In terms of LPIPS, BRC and BRC-CDM also demonstrate superior perceptual quality. Other methods tend to lose spatial texture and fine details, particularly at low bit rates. In contrast, BRC provides sharper texture reconstruction, while BRC-CDM achieves the highest perceptual similarity among all methods. This highlights the strength of the compression-driven generation strategy in restoring fine-grained spatial structure and maintaining visual coherence.

Together, these results confirm that BRC and BRC-CDM offer a better balance between spectral preservation and perceptual quality. BRC-CDM, in particular, shows strong potential in high-fidelity hyperspectral compression tasks.

E. Ablation Study on Network Architecture

In order to demonstrate the ablation study of the network architecture, we conduct a comparative analysis of two distinct scenarios.

1) BRC-W/O RCC (Without Residual Correction Compression): This variant uses only the SGGC module to process the reference band. A predicted image of the current band is generated through volumetric rendering. No information from the actual current band is used. The predicted image is directly used as the final output.

2) BRC (Spectral Residual Compression + SGGC + Residual Correction Compression): The proposed full model first generates a predicted image using the SGGC module. Then, it models and compresses the residual. This forms a collaborative prediction-residual compression mechanism. The design achieves a better balance between compression efficiency and reconstruction quality.

The results indicate that the proposed model demonstrates superior coding efficiency compared to the variant that lacks residual correction. It is also noteworthy that, without interaction with the actual band, the generated image mainly depends on the reference band features. We select the Washington

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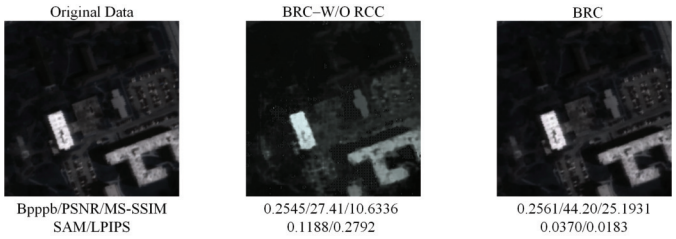


Fig. 8. Visualization of Washington dc dataset on ablation study.

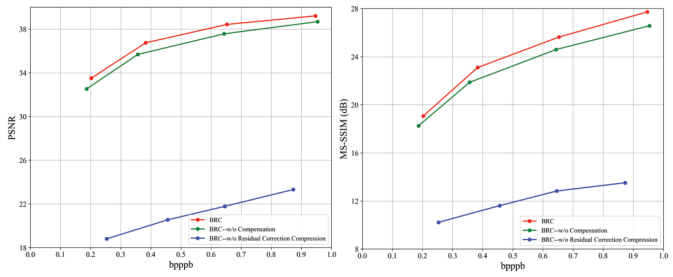


Fig. 9. Rate–distortion performance on ablation study.

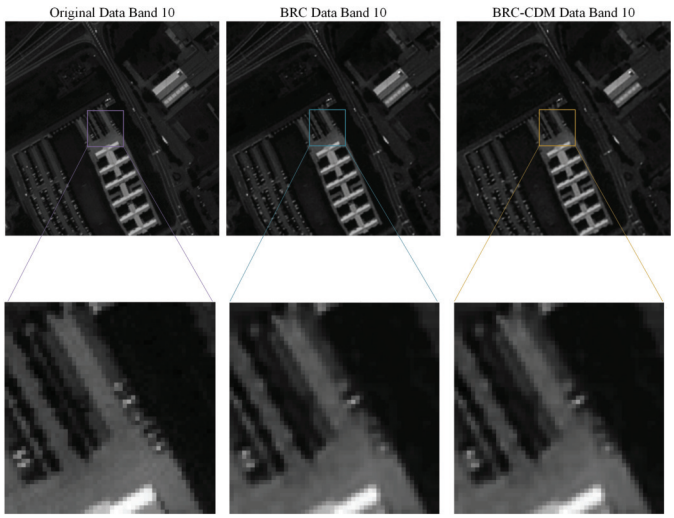


Fig. 10. Visualization of selected areas.

dc dataset for visualization of ablation study. The results are presented in Fig. 8.

In ablation studies, the base model using only reference-based prediction shows strong limitations. The errors are especially visible at low bit rates. The biresidual structure compensates for the unmodeled components and improves both structural accuracy and perceptual quality. The improvement results from addressing prediction limitations, not from convergence instability or architectural flaws.

Furthermore, we present their rate–distortion performance as assessed on the comprehensive dataset, as illustrated in Fig. 9. In order to illustrate the model's performance across the spectral dimension, the Pavia University dataset has been chosen for analysis. A spatial region of size 50 × 50 pixels is extracted from the image for analysis, which is shown as in Fig. 10.

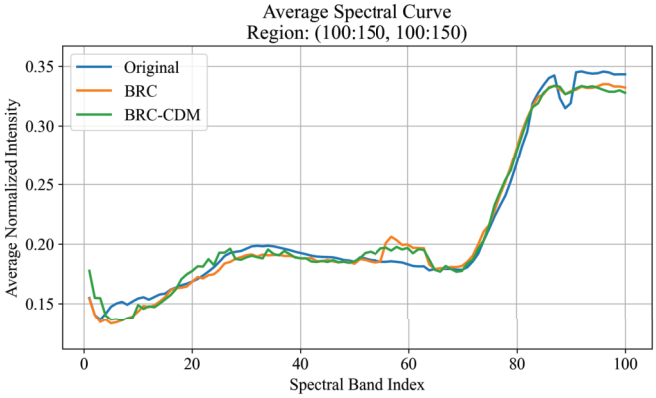


Fig. 11. Spectral curve visualization.

Within this region, the average pixel intensity is computed for each spectral band. These average values are then used to generate the spectral reflectance curves. This visualization allows for a direct comparison of spectral consistency between the original data and the reconstructed outputs from different models, which is shown as in Fig. 11.

The proposed BRC and BRC-CDM methods generate spectral curves that closely follow the ground-truth profile. The reconstructed curves align well with the reference across the entire spectral range, indicating accurate preservation of reflectance values at each band.

In particular, BRC-CDM demonstrates strong spectral consistency and significantly reduces spectral distortion. The method effectively maintains the shape and continuity of spectral signatures, even under compression. This confirms its ability to reconstruct hyperspectral data with high fidelity in the spectral domain.

F. Comparison of Visualization Results

Four images are randomly selected from four distinct test datasets for the purpose of reconstruction and subsequent visualization. To ensure comparability, the spatial resolution of all visualized images is uniformly set to 256 × 256, with the specific results shown in Figs. 12–15.

Traditional codecs such as BPG [11] and JPEG2000 [9] often introduce noticeable block artifacts and blur, especially in regions with fine textures. These distortions reduce image sharpness and degrade structural detail. Similarly, images processed using PCA [34] show considerable blurring and a loss of detail.

In these figures, while Mijares i Verdú et al. [24] and the Context [17] methods effectively compress overall information, they struggle to preserve fine details and consume more bits than BRC and BRC-CDM. The CDC model [29], on the other hand, results in overly smooth images with significant detail loss. Methods such as FHNerf [23] and Cheng et al. [35] retain some detail at similar bit rates, but still fall short of BRC and BRC-CDM, which highlights the latter's advantages in visual perception and detail reconstruction.

Meanwhile, JSCIFI-F [25] and ARM-Net [26] show visible distortions in certain regions. These distortions often appear

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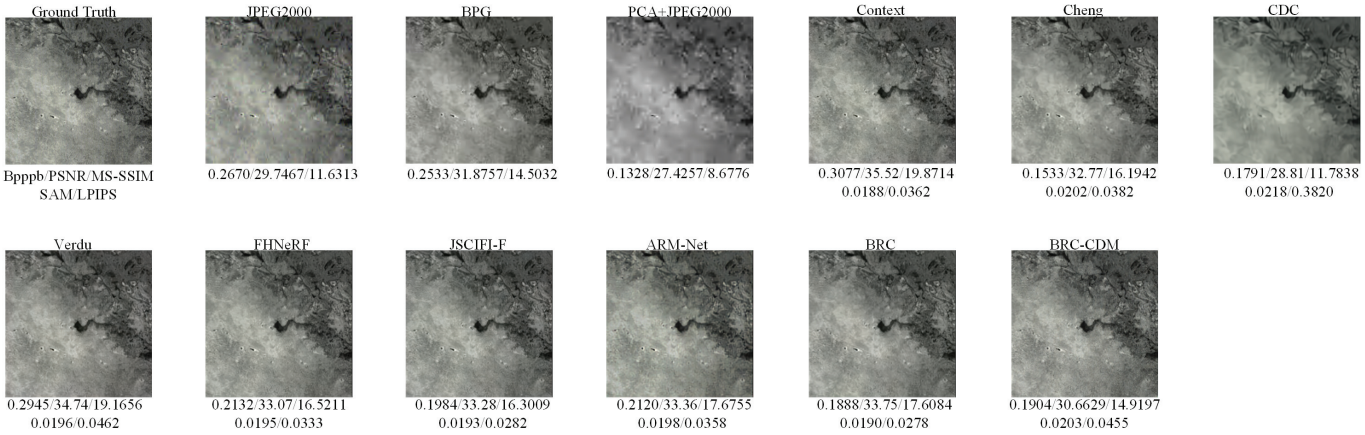


Fig. 12. Visualization of Botswana dataset.

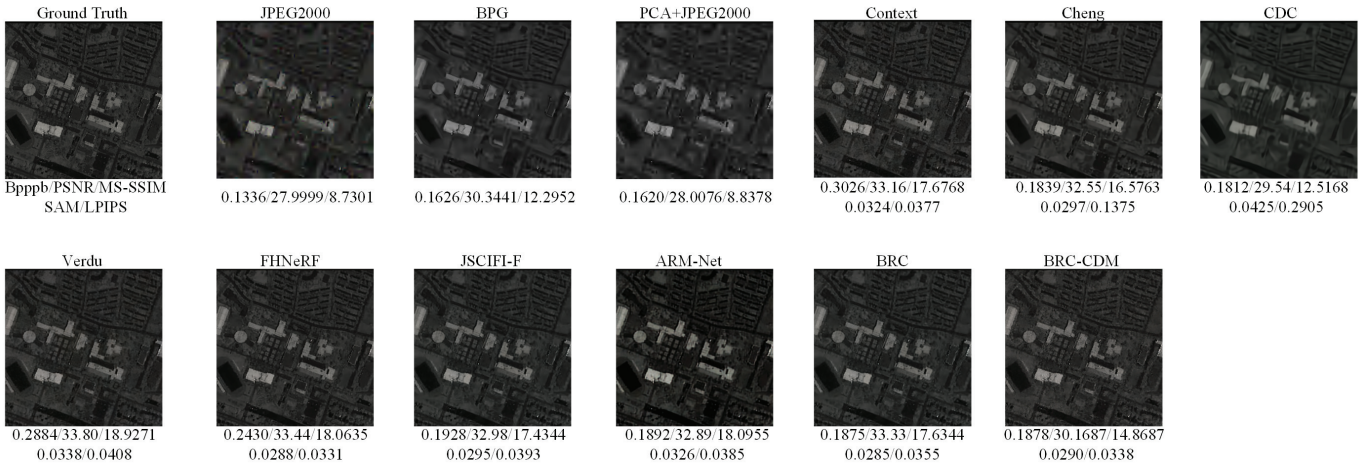


Fig. 13. Visualization of KSC dataset.

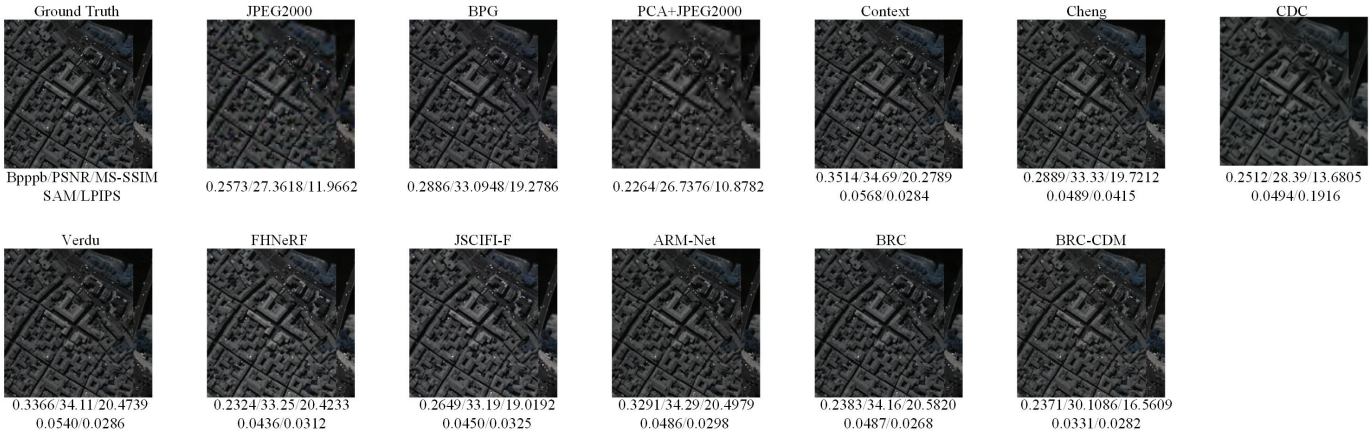


Fig. 14. Visualization of Pavia dataset.

as blurred edges and loss of texture detail. These methods achieve lower reconstruction quality compared to the proposed BRC. Their performance in LPIPS and SAM also remains consistently inferior to both BRC and BRC-CDM.

Although Mijares i Verdú et al. [24] and the Context [17] method use the most bpppb, their results are comparable to BRC, while SAM and LPIPS perform worse than BRC-CDM. At low bit rates, traditional algorithms lose much detail. The method of Cheng et al. [35] yields better PSNR results than BRC-CDM. However, it fails to preserve fine details. This highlights the limitation of relying solely on PSNR and MS-SSIM for evaluating overall image quality.

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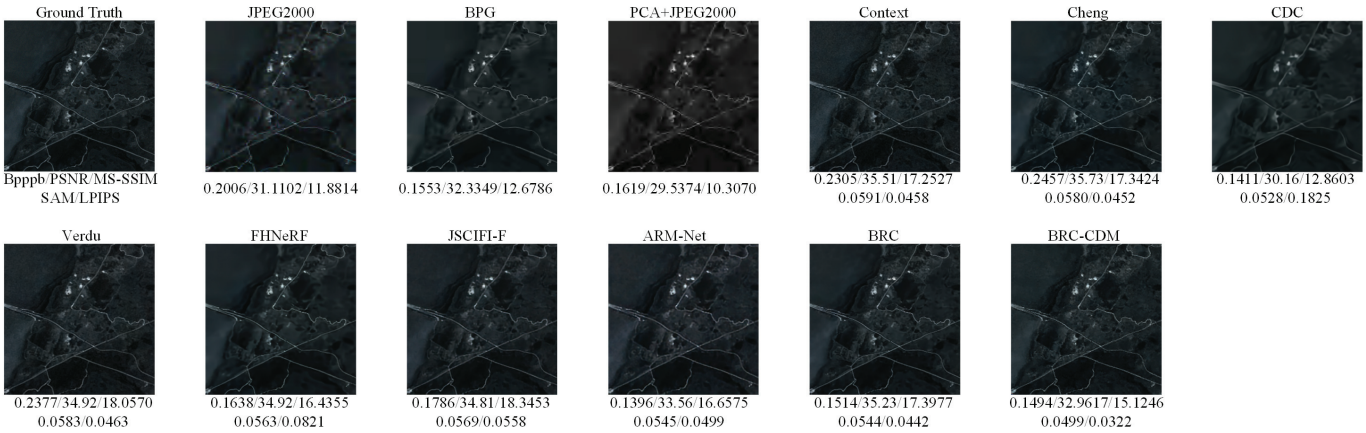


Fig. 15. Visualization of Houston dataset.

TABLE III

CLASSIFICATION ACCURACY OF DIFFERENT RECONSTRUCTIONS OBTAINED FROM COMPARED COMPRESSION METHODS AND THE PROPOSED BRCNET AND BRC-CDM FOR THE PAVIA CENTER DATASET AT DIFFERENT BPPPB

Bpppb	Class	Original	Context [17]	Cheng [35]	Verdu [24]	FHNerf [23]	JSCIFI-F [25]	ARM-Net [26]	BRC	BRC-CDM
0.20	OA	0.9765	0.9521	0.9547	0.9543	0.9553	0.9552	0.9554	0.9558	0.9560
0.55	OA	0.9765	0.9567	0.9583	0.9569	0.9583	0.9630	0.9651	0.9707	0.9718

TABLE IV

COMPARISON OF COMPRESSION METHODS IN TERMS OF MODEL SIZE, SPEED, AND COMPLEXITY

Method	Parameters (M)	FPS	FLOPs (G)	Encoding Time (s)	Decoding Time (s)
Jpeg2000 [9]	–	–	–	1.15	0.89
PCA+jpeg2000 [34]	–	–	–	1.55	1.02
BPG [11]	–	–	–	1.09	0.75
Context (2018) [17]	10.5	52.15	25.35	0.35	0.049
Cheng (2020) [35]	9.7	57.27	27.49	0.312	0.025
CNN_HSI (2021) [43]	156.78	55.65	213.51	0.3687	0.0798
SSCNet (2022) [44]	198.92	51.37	265.49	0.4821	0.1812
CDC (2023) [29]	153.85	56.25	117.87	0.22	1.04
Verdu (2023) [24]	5.08	107.14	12.34	0.248	0.023
CENet (2023) [22]	35.40	6.00	144.00	0.4500	0.1300
HCCNet (2023) [45]	41.74	4.31	146.00	0.4311	0.2114
FHNerf (2024) [23]	6.2	80.27	13.2	0.15	0.013
JSCIFI-F (2025) [25]	22.17	68.34	29.57	0.4561	0.2396
ARM-Net (2025) [26]	42.15	5.96	398.00	1.841	0.322
BRC (Ours)	21.89	52.64	26.85	0.059	0.003
BRC-CDM (Ours)	60.25	19.89	85.46	0.063	0.112

The CDC method [29] suffers from overly smooth detail processing, and FHNeRF [23] similarly fails to preserve fine details and edge information at low bit rates.

G. Effect on Classification Results

To enhance model generalization, two classification datasets are introduced. A land-cover classification task is conducted on the Pavia Center dataset to verify the model's effectiveness. We chose to use the 2-D convolution method proposed by Lee and Kwon [41]. PCA dimensionality reduction is not used, as we performed specific preprocessing on the datasets. To present our final classification results, we decide to use the overall accuracy (OA) metric. The findings of the comparative analysis are presented in Table III.

The following conclusions can be drawn from the results: first, the proposed compression network shows good performance under low bit rate conditions. Second, even at high bit rates, the model still outperforms other compression methods. Furthermore, when the difference in PSNR is negligible, the improvement in other metrics (e.g., SAM and LPIPS) positively affects the classification accuracy.

H. Model Complexity

Table IV demonstrates that the proposed BRC model maintains an effective tradeoff between network complexity and compression efficiency. It maintains a moderate parameter count and low computational cost. It also delivers high inference speed, as indicated by its superior frames per second (FPS). The BRC model employs a biresidual architecture, which introduces a relatively higher parameter count compared to lightweight designs. Nevertheless, it consistently outperforms all baseline methods in compression quality. This confirms the efficiency of its spectral residual modeling.

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The extended BRC-CDM model also shows competitive performance. It integrates a diffusion-based decoder yet retains fewer parameters than other diffusion-based models. Although more complex than BRC, BRC-CDM achieves strong results in both spectral fidelity and perceptual quality.

V. Discussion

The proposed BRC-CDM framework integrates conditional diffusion modeling with residual-based compression to improve HSI reconstruction. A compact latent representation is generated using a BRC structure. This representation encodes key spectral priors and spatial structures. This latent condition functions as a guiding parameter for the diffusion model throughout the reverse denoising process. The model systematically reconstructs the target band from Gaussian noise, utilizing this condition to direct the reconstruction.

This structure enables accurate recovery of structural contours and fine textures. At high bit rates, the condition provides sufficient guidance, allowing the model to maintain strong spectral fidelity and perceptual quality. At low bit rates, however, the compressed representation carries less information. The diffusion process becomes increasingly dependent on its internal prior, which is learned from the training dataset. This may lead to the generation of hallucinated textures or structural artifacts, especially when the current input deviates from the training distribution. These errors may degrade reconstruction quality and limit the reliability of results in downstream tasks such as classification, detection, or analysis.

From a deployment perspective, CDM is only involved in the decoding phase. It does not affect the runtime or efficiency of the encoder. This design makes the framework suitable for real-world scenarios, where encoding may occur on edge devices with limited resources, and decoding can be handled on more capable platforms such as ground stations, cloud servers, or remote sensing data centers. In addition, the model can support flexible deployment strategies. CDM can be pretrained and fine-tuned for specific datasets. It can also be selectively applied to critical spectral bands to reduce inference cost without compromising reconstruction quality.

Future improvements may include enhanced conditional priors or multimodal conditioning strategies. These could improve robustness under low bit rate settings and reduce reliance on implicit generative priors. Such enhancements will further improve the applicability of BRC-CDM in operational HSI compression tasks.

VI. Conclusion

In this study, we introduced the BRC-CDM framework to address the challenges of HSI compression and reconstruction, and demonstrated its innovation and effectiveness. The core advantage of the BRC-CDM framework lies in its band selection strategy, which employs distinct bands for the input HSI and the reference image. This design leverages the interband correlations, significantly enhancing both the efficiency and quality of the compression and reconstruction processes.

We start by extracting the residuals between the previous band and the current band using an encoder–decoder network, which captures details that are difficult to predict directly. These residuals are then integrated with the outputs of the multiscale feature pyramid, which captures image features at different scales.

Moreover, the accuracy of the reconstructed image is maintained through the computation of discrepancies between the predicted and actual bands. These prediction errors are rectified by incorporating the residuals into the predicted bands. To further improve the quality of the reconstructed image, a CDM is integrated as the concluding phase of the image generation process. Quantitative and visual analysis of nine different HSI datasets demonstrates that the BRC framework exploits band correlations to improve compression performance.

In future work, we plan to integrate task-specific features into the compression process to optimize intermediate variables more effectively. Through the optimization of the band selection strategy and the refinement of the diffusion model, we aim to improve the accuracy of image reconstruction as well as the overall performance of the system. These enhancements are expected to advance the capabilities of HSI processing technology and expand its applicability in practical contexts.

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